



Mahindra University, Hyderabad
École Centrale School of Engineering
MINOR II Examinations, Fall semester 2025 (2023 Batch)
Program: B. Tech. (Common to CM and NT)
Year: III Semester: I
Subject: Computational Methods for PDE (MA 3115)

Date: 30/10/2025
Time Duration: 1.5 hours

Time: 10:00 -11:30 AM
Max. Marks: 15

Instructions:

1. All the answers that belong to a particular question should be answered in one place in your answer booklet.
2. Scientific calculators are permitted.

Q 1:

Marks: 05

Find the Fourier series representation of a periodic function $\phi(x)$ with period 2π ,

$$\phi(x) = \begin{cases} \pi + x, & -\pi < x \leq 0 \\ \pi - x, & 0 < x \leq \pi. \end{cases}$$

Q 2:

Marks: 05

Find the solution of 1D wave equation using method of separation of variables,

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < 1, t > 0; u(0, t) = 0, u(1, t) = 0, t > 0;$$

$$u(x, 0) = \frac{1}{8} \sin(\pi x), \quad 0 < x < 1, \quad \frac{\partial u}{\partial t}(x, 0) = 0, \quad 0 < x < 1.$$

Q 3:

Marks: 05

Derive CTCS scheme and perform two iterations to solve the 1D wave equation given in

Q2, with $\Delta x = \frac{1}{4}$ and $r = \frac{1}{2}$.